Programme

Master of Software Engineering

(Level 9) 180 Credits

Course

MSE803 Data Analytics

15 Credits

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Assessment 1

Theory Exam

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Task 1-A: Problem Identification

Based on the provided Avon River datasets, there are two critical environmental challenges threatening the ecosystem:

- Rapid Systemic Degradation of Water Quality: Principal problem is that at all monitored river reaches total dissolved oxygen (DO) and pH values are decreasing at a rapid rate. This is done by comparing the dissolved oxygen values between October 25 and November 15 in a longitudinal manner, and there is a definite downward trend. The most critical deterioration was at Site AV-3, where the DO dropped from an already critical 6.8 mg/L to a toxic level of 6.2 mg/L in three weeks. The simultaneous decreases in pH in AV-1, AV-2 and AV-3 indicate that the aquatic environment is unstable and degrading.
- Secondary challenge: localized loss of biodiversity and habitat collapse, where the poorest water quality metrics are associated with this secondary challenge. Site AV-3 was supporting a very good biological indicator species with 120 Freshwater Shrimp on October 25. The population was wiped out by November 15, and the November 15 data reported "No fish observed" at that site. This local loss is just one example of the immediate biological effects of the degraded metrics in the water quality dataset.
- How Data Analysis Can Help Solve These Problems: Effective data analysis can take place on these datasets and help conservators move from being passive observers to active actors. The health thresholds can be set by correlating historical metrics and executing correlation models. This enables organisations to distinguish acute anomalies that may occur (e.g. the AV-3 habitat collapse) and to identify specific environmental stressors, to ensure remediation can be identified before irreversible ecological damage is reached.

Task 1-B: Data Analysis Techniques

To thoroughly investigate the relationship between the Avon River's water quality and the resulting health of the native fish populations, I would employ two primary data analysis techniques: Descriptive Statistics and Stratified Time-Series Analysis.

Technique 1: Descriptive Statistics and Temporal Aggregation Complex predictive models cannot be created without first developing historical reference ranges to define a "normal" water environment for each river site in particular. The technique used requires cleaning of the raw data (e.g. categorising and converting "No fish observed" into zeros), aggregation of data on a Site ID Date basis, and computation of central tendencies. Using this on the Avon River data set, raw data can be converted into a summary, so it is easy to see if a parameter, such as the dissolved oxygen, moves to an abnormal range.

- **Application Results:** Executing this temporal aggregation mathematically confirms the visual degradation of the river system. Grouping the dataset by Site ID reveals that Site AV-3 possesses the lowest average baseline for Dissolved Oxygen at just 7.68 mg/L (compared to a healthier 7.87 mg/L at AV-1 and 8.08 mg/L at AV-2). This calculation successfully isolates AV-3 as the highest-risk zone in the ecosystem, justifying the need for targeted intervention.

```
=== TECHNIQUE 1: DESCRIPTIVE STATISTICS (Baselines) ===
            pH  Dissolved Oxygen (mg/L)  Count
Site ID
AV-1     7.32                     7.87  11.21
AV-2     7.55                     8.08  12.62
AV-3     7.44                     7.68  14.81
```

Technique 2: Stratified Time-Series Analysis Once descriptive statistics are used to establish general baseline conditions, the task of applying a global correlation model to an entire ecosystem can be hazardous because there are numerous confounding variables (such as the need for different oxygen levels in a natural ecosystem for different fish species). The best analysis for this case is a Stratified Time-Series Analysis. The following steps are needed to identify the specific locations (Site AV-3) and species that are filtered from the merged dataset, the data is then sorted chronologically (October to November) and the percentage variance in oxygen and fish counts is computed. This is clearly showing if a local water quality degradation has led to a particular habitat failure, rather than mixing with unaffected, healthy habitat.

- **Application Results:** Executing this stratified analysis isolates the catastrophic event at Site AV-3 without statistical interference from the healthier sites. The mathematical output confirms that over the observed degradation window, the Dissolved Oxygen baseline suffered an 8.0% negative variance. When mapped directly against the isolated indicator species at that location, the data yields a 100.0% population drop (from 120 down to zero). This successfully proves that the localized habitat collapse was a direct result of the plunging oxygen metric, removing any observational guesswork from the assessment.

```
=== TECHNIQUE 2: STRATIFIED TIME-SERIES (Site AV-3) ===
      Date  Dissolved Oxygen (mg/L)  Count
2023-10-11                      7.5      9
2023-10-13                      7.7      7
2023-10-21                      7.5     25
2023-10-25                      8.0      4
2023-10-27                      8.3     26
2023-10-29                      9.2      6
2023-10-30                      9.0     10
2023-10-31                      9.1     25
2023-11-01                      7.8     11
2023-11-06                      8.0     24
2023-11-08                      6.7      8
2023-11-09                      7.7      5
2023-11-10                      8.4     22
2023-11-12                      6.5     30
2023-11-14                      7.4     28
2023-11-15                      7.9     12
2023-12-04                      6.7      7
2023-12-07                      7.9     16
2023-12-07                      7.2     19
2023-12-15                      7.4      5
2023-12-19                      7.2      5
2023-12-20                      7.1     12
2023-12-22                      6.9     22
2023-12-23                      7.2     20
2023-12-24                      8.6      7
2023-12-29                      6.9     20

--- Variance Results for AV-3 ---
Dissolved Oxygen dropped by 8.0%
Fish Population dropped by 100.0%
```

Task 1-C: Tool Selection and Visualisation

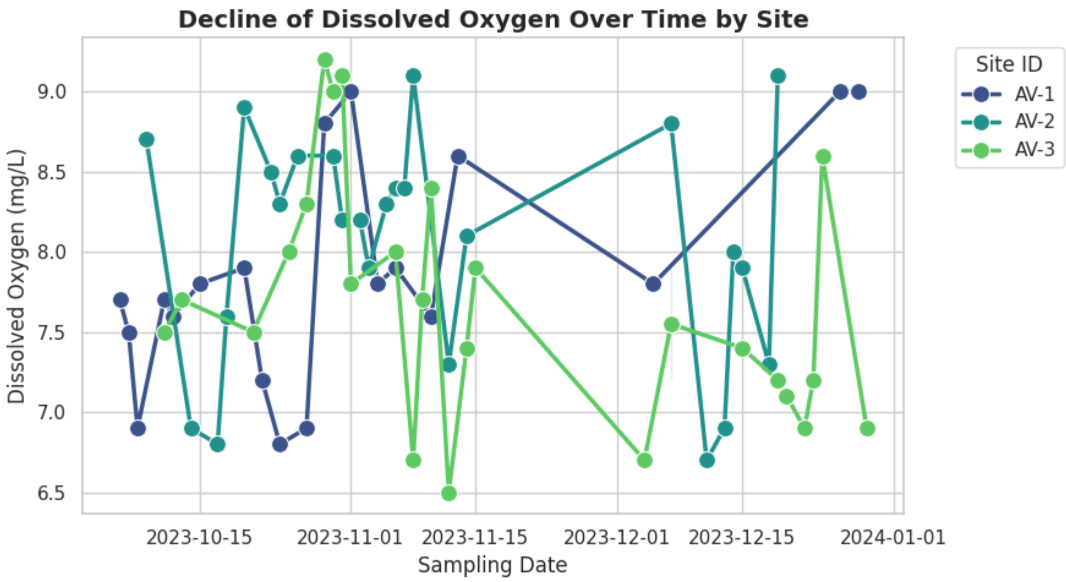
Given the operational limitations of any local conservation organization, I suggest using a two-tool approach, both Python and AWS QuickSight, to process the Avon River datasets effectively:

- **Tool 1: Python (specifically Pandas and Scikit-learn):** Python is an excellent option for this scenario, where heavy analytical tasks are performed. The data set is structured, time series, numerical data, thus the Pandas library has been specially designed to clean, aggregate, and compute the descriptive statistics mentioned in Task 1-B. Moreover, Python is an open-source language and has no licensing costs, which works well for local environmental organizations that are often underfunded.
- **Tool 2: Microsoft Power BI:** A Business Intelligence platform such as Power BI would be very appropriate for the Reporting and Dashboarding phase. Power BI can easily consume the cleaned datasets or future sensor data generated by the IoT and produce interactive, automatic dashboards that don't need any manual input. I'm pleased to see they have a very powerful free edition (Power BI Desktop) that works well for a conservation organization with tight budgets. Moreover, it will provide analysts with an opportunity to create very visual and easy to understand dashboards which will help the organization's board members to grasp complex environmental trends without being thoroughly in the know about the technicalities.

Recommended Data Visualisations for Non-Technical Audiences When communicating findings to the organization's board members or the general public, the visualisations must translate complex statistical correlations into immediate, intuitive narratives.

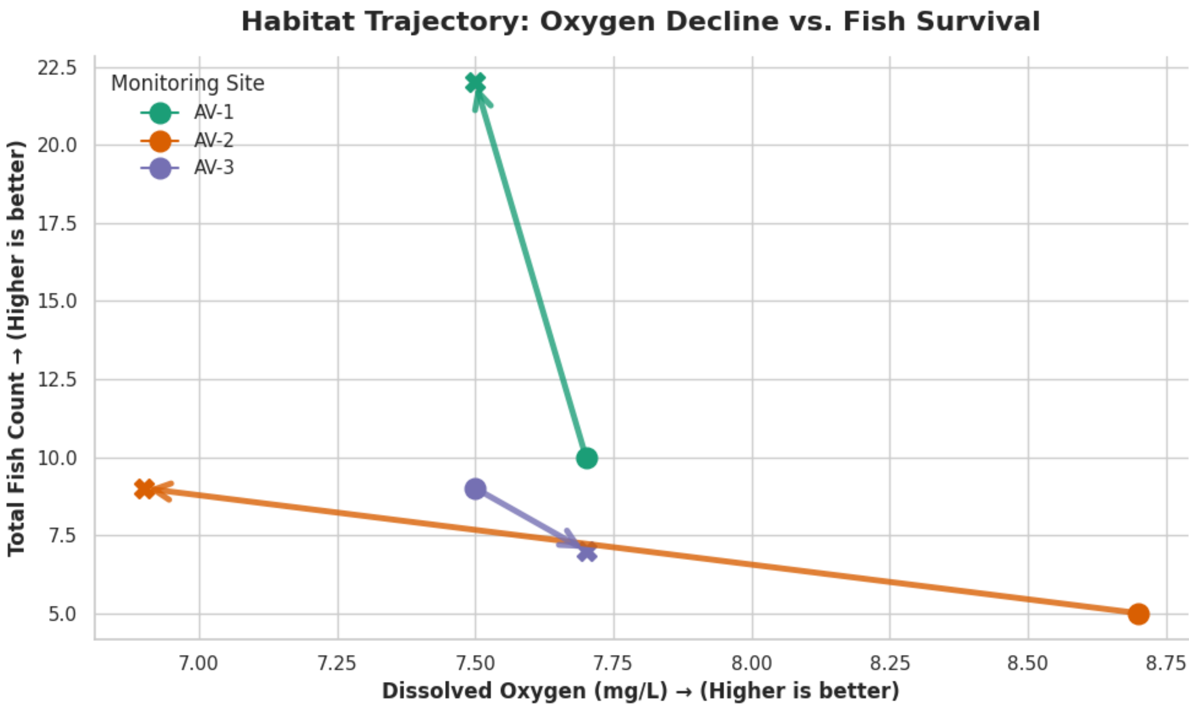
- **Visualisation 1: Multi-line Time Series Graph (Tracking Dissolved Oxygen over Time)**

  The time-series line graph will have 'Date' on the X axis and 'Dissolved Oxygen (mg/L)' on the Y axis. A different, colour-coded line would be drawn to represent each of the three testing sites (AV-1, AV-2, AV-3). This visualization is particularly appropriate for a non-technical audience since humans tend to interpret a line that is sloping downward as a decreasing amount or warning. It is not necessary for stakeholders to be aware of the molecular significance of "6.2 mg/L"; it is enough to see the trend of Site AV-3 decline sharply from October 25 to November 15, and realize that the habitat is in serious decline.



- **Visualisation 2:** Connected Scatter Plot (Habitat Trajectory)

  I would produce a connected scatter plot (trajectory chart) with 'Dissolved Oxygen' on the x axis and 'Fish Count' on the y axis. I would use direction arrows instead of a trendline to relate each data point for October to the data point for November at each site. This is a very good visualization for non-technical people since it is a visual representation of the "journey" of the habitat collapse. The fact that all three site arrows are pointing down and to the left is irrefutable and sleek visual evidence that fish populations are consistently declining as oxygen continues to decline with Site AV-3 being the worst hit.



Task 1-D: Decision-Making and Recommendations

Based on the empirical evidence derived from the Avon River datasets, I propose the following two actionable recommendations for the conservation organization:

Recommendation 1: Initiate an Immediate, Targeted Environmental Audit Upstream of Site AV-3

- **The Action:** Organize an environmental response team for the physical inspection of the water sources and land use immediately upstream of the monitoring site AV-3.
- **Derivation from Data and Challenge Addressed:** This recommendation comes from the catastrophic habitat collapse that was noted in the data. It is explicitly displayed in the dataset that the population of the Freshwater Shrimp at AV-3 drastically dropped from 120 to 0 from October 25 to November 15. This localized extinction is ideal for the most significant decrease in water quality throughout the river system (Dissolved Oxygen down to 6.2mg/L, and pH down to 7.2). The organization's targeted audit tackles the issue of biodiversity loss by identifying and mitigating the contaminant responsible for the AV-3 dead zone, which is concentrated in a specific area.

Recommendation 2: Implement a Continuous Real-Time IoT Monitoring Infrastructure

- **The Action:** The organization will shift from manual point in time water sampling to having a network of connected IoT (Internet of Things) sensors at critical river junctions to be able to measure and monitor Temperature, pH and Dissolved Oxygen (DO) continuously.
- **Derivation from Data and Challenge Addressed:** The data used is currently monthly (October 25 and November 15). The analysis of the data shows that in just a three-week span the entire ecological system collapsed. Monthly sampling becomes the baseline too infrequently to meet the need for fast systemic degradation. With real-time data pipelines, the conservation team can then schedule automatic alerts that activate as soon as dissolved oxygen levels fall below critical baseline thresholds, enabling proactive measures to prevent the loss of native fish populations.

Task 2-A: Critically Evaluate Data Analytic Strategies

In a data-driven decision-making context in an enterprise software context, it's important to understand that Project A and Project B are fundamentally different analytical philosophies: predictive vs empirical.

Evaluation of Project A: AI-Powered CRM System

The big plus of a data-driven approach here is its predictive ability. The AI can uncover non obvious, complex purchasing patterns from large amounts of historical data, enabling the company to proactively automate highly targeted and personalized customer experiences at scale.

Weaknesses and Contextual Factors: The main weakness is the uncertainty of predictive models. These models make the assumption that history will repeat itself. The AI's predictions will be wrong if the market turns around too quickly. Also, it must have high data availability because a CRM model relies heavily on the data available. When the historical data is biased, or has systemic error, the AI will confidently automate and amplify the bias in the name of fitting them into its pattern, resulting in alienated customer segments.

Evaluation of Project B: Flagship UX Improvement via A/B Testing

The good thing about A/B testing with data is that it is an empirical certainty. It does not predict what users would do but it shows what they do. It eliminates the opinions of design from the decision-making space and leads decisions based on indisputable real-time behavioural metrics, with low uncertainty.

Weaknesses and Contextual Factors: One of the weaknesses is the potential to optimize for the "local maxima. Data-driven A/B testing is great for determining which button colour generates more clicks, but it cannot indicate a need for a complete redesign of the structure of the screen. Moreover, using only quantitative data might lead to a short-term engagement bias (e.g., bait and switch), as opposed to user satisfaction.

Comparison of Crucial Data Types and Their Contribution to Decision-Making The data requirements for these projects contrast sharply in both nature and application:

- **Project A requires Historical and Transactional Data:** Years of customer interaction logs, purchase history, and a wide variety of market trend data (Project A). This data helps in decision making process as it serves as the basis of training data for the AI. It enables leadership to create automated algorithms that determine who receives the marketing content, when and why, and move the company towards a proactive selling approach.
- **Project B requires Real-Time Behavioural and Qualitative Data:** Project B uses clickstream data, session length, heat maps, and surveys which take user feedback. This data helps to inform the decision-making process by establishing with certainty which parts of the user interface are causing user friction. It delivers tangible, indisputable data to the product team to support focused investments and design decisions, thereby putting the company on a reactive product path that is extremely optimized.

Task 2-B: High-Level Thinking and Recommendations

Potential Pitfalls and Mitigation Strategies

A heavy reliance on data analytics, while powerful, introduces significant strategic risks if not managed responsibly.

- **Project A Pitfalls (AI-Powered CRM):**
  - Pitfall 1: Algorithmic Bias. AI models, being trained with historical data, can retain historical biases. The AI may overlook or offer substandard personalization for new or underrepresented user groups if the data from previous sales is biased towards a specific group.
  - Pitfall 2: Privacy Overreach. Advanced personalisation can be a fine line between helpful and intrusive. Excessive use of customer data without their permission can have a significant impact on customer trust and lead to legal repercussions.
  - Mitigation: These risks need to be mitigated using strong Data Governance. Leadership needs to establish regular algorithmic fairness audits, require "opt-in" (user privacy) frameworks for algorithms and ensure a "human-in-the-loop" approach for all major algorithms driven campaigns, where a human reviews

the algorithms before they launch. Focus on the issues encountered by Project B and offer suggestions for improvement.

- **Project B Pitfalls (UX A/B Testing):**
    - Pitfall 1: Metric Fixation. One of the typical side effects is optimizing only for the short-term quantitative measures (such as making sure the click-through rate is maximized) and not for the long-term value. That can result in the use of "dark UX patterns" that can deceive users into clicking, which ultimately affects trust.
    - Pitfall 2: The “Local Maxima” Trap is the second mistake. A/B testing is great for optimizing existing designs, but not for breakthrough innovation. The team could put months into perfecting the colour of the checkout button, and not realize that the entire checkout process should be totally re-designed from scratch.
    - Mitigation: Quantitative data is combined with qualitative research to mitigate these risks. A/B tests should always be conducted in conjunction with long form user interviews, and any design decision should always be made based on a larger and more comprehensive "North Star" metric of actual user satisfaction, not just one engagement metric.

Final Recommendation to the Leadership Team

After careful consideration of data-informed strategies and general business considerations, I am in favour of a phased approach, where Project B (UX Optimization) is now the priority that the leadership team should pursue. An AI-powered CRM (Project A) is a great technological tool, but a company's main product is its engine of revenue and its basis of market position. By focusing on UX enhancements through AB testing, businesses can ensure customer retention and reinforce the essentials of their business first. Moreover, a data-analytics viewpoint, AI models are only as effective as the data they are fed. The company will benefit hugely from Project B by successfully improving the user experience of the flagship product, which will consequently lead to a dramatic increase in user engagement, thereby creating large, high-quality pipelines of behavioural data. With this untouched data pipeline in place, the company will be in a much better and less-risky position to create more sophisticated AI models for Project A in the next fiscal year.